Production and network coverage

Production, simulated coverage, and projected network-mean error

a Production flow

Suppliers · per production lot

Slit sheet and carrier
Luxel builds

Metal foils: Zr, Al, In
Luxel bonds them to the sheet

FUV filters
Catalogue vendor, per order

Detectors
Sensor vendor, per wafer lot

Boards
Fabrication and assembly, per panel

Carthage · every unit

Optical head
Fit partitions, set the 5.0 mm standoff, clamp FUV filters

Unit assembly
Boards, heat path, housing

Unit test
Flat field, dark current against temperature, timing, vibration

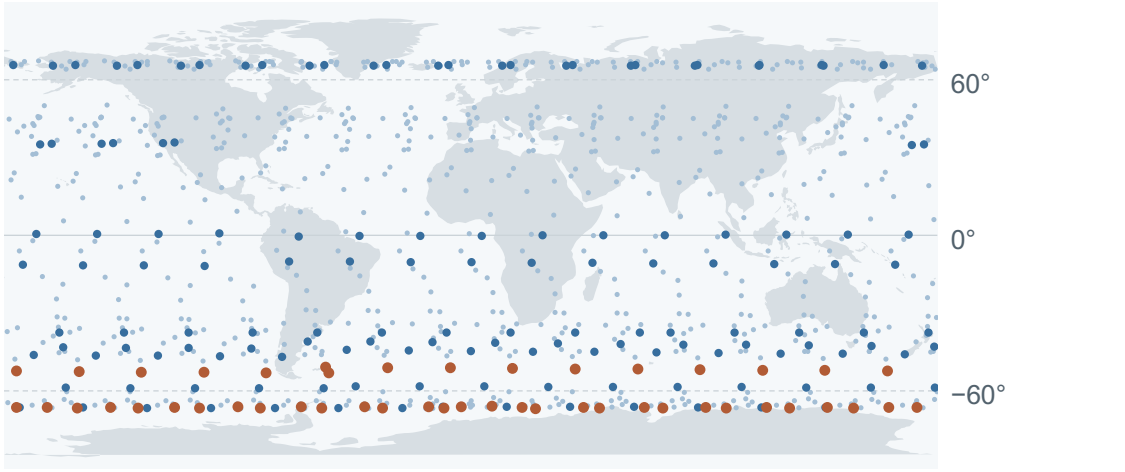
Calibration

Ground calibration
Angular scan: every slit plate
NIST SURF III, 5–175 nm: one unit per 20-unit lot

Launch on a host

In flight, every occultation
Reference rows give I_0;
units are cross-compared

b Simulated occultations, 1 January 2026

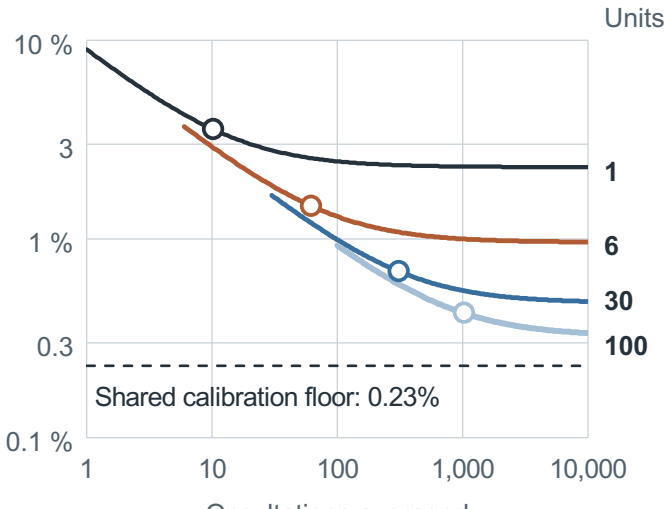


● Units 1–6: 47 total ● Units 7–30: 189 total ● Units 31–100: 940 total

Model: independent 8.7% random errors per occultation.
Assumed 2.3% unit offset; 1% of its variance is shared.
Balanced sampling of a common density; physical variability excluded.

c Random + calibration error

○ One day at annual-average rate



Mean mass density at 200 km, severe storm

Figure 5. (a) Batch production and unit testing. (b) Sunset rays at 200 km; hosts: 42 in the 53° shell, 25 sun-synchronous, 33 other; counts cumulative. (c) Network-mean error; page 2 bias adds. Markers: one day at 10.33 occultations per unit. With 30 % of each offset shared per 20-unit lot, 100 units reach 0.64 %.